# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

**MG1DSCBTS100 – NEW BIOLOGY FOR THE NEW CENTURY**

**Duration: 2 hrs** **Maximum Marks: 70**

*Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.*

## Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries **1** mark

1. Which kingdom includes organisms that are primarily unicellular and lack a nucleus? **[understand] [2]**

a) Plantae b) Animalia c) Fungi d) Monera

2. The enzyme blocked in Flavr Savr Tomato is **[Apply] [7]**

a) Urease. b) Polygalactouronase. c) Cellulase. d) Lipase

3. Haematoxylin and Eosin are [Remember] [3]

a) Enzymes. b) Antibodies c) Dyes d) Dyes

4. What type of biomolecule is primarily responsible for storing genetic information? **[Remember] [3]**

a) Proteins b) Lipids c) Nucleic acids d) Carbohydrates

5. How do viruses differ from living organisms? **[Analyse] [1]**

a) They can reproduce independently b) They contain genetic material

c) They cannot carry out metabolic processes on their own d) They have a cellular structure

6. What is the function of lysosomes in a cell? **[Understand] [5]**

a) Energy production b) Digestion and waste removal c) Protein synthesis d) Photosynthesis

7. What does the genetic code primarily consist of? **[Remember] [4]**

a) Amino acids b) Codons c) Nucleotides d) Proteins

8. What is the phase of the cell cycle where the cell grows and prepares for division? **[Remember] [1]**

a) G1 phase b) S phase c) G2 phase d) M phase

9. In the context of evolution, what does the term "fitness" refer to? **[Understand] [7]**

a) Physical strength of an organism b) Reproductive success of an organism

c) Overall health of an organism d) Ability to adapt quickly

10. What are the fundamental principles of biology that enable organisms to exist? **[Remember] [1]**

a) Homeostasis b) Adaptation c) Reproduction d) All of the above

**[1 x 10 = 10]**

## Part B

Short Essay Type Questions

Answer **6** Questions

Each question carries **5 marks**

11. What are the two types of endoplasmic reticulum in eukaryotic cells, and how do they differ in function?

**[Understand] [2]**

12. Provide one example each of an organism from the Monera and Protista kingdoms. **[Remember] [2]**
13. What are the main phases of the cell cycle? **[Remember] [1]**
14. Describe the differences in genetic composition between the daughter cells produced by mitosis and meiosis. **[Analyse] [5]**
15. Differentiate between light microscopy and fluoroscent microscopy**.** **[Analyse] [7]**
16. What are restriction endonucleases and what are they used for? **[Apply] [4,7]**
17. What is HGP? **[Remember] [1,5,7]**
18. List out the steps in gene cloning. **[Understand] [4,7]**

**[5 x 6 = 30]**

## Part C

Essay Type Questions

Answer **2** Questions

Each question carries **15 marks**

19. Describe four characteristics that are common to all living organisms **[Understand] [1]**
20. Discuss the principles of inheritance as proposed by Gregor Mendel and how they apply to modern genetics. **[Understand] [4]**
21. Identify and describe three common misconceptions about evolution and clarify them. **[Analyse][6]**
22. Describe in detail on NMR and X-ray crystallography**.** **[Apply] [3,7]**

**[15 x 2 = 30]**